# DIFNAL - K

**(Diclofenac Potassium Tablets 50 mg)**

## COMPOSITION

Each film coated tablets contains:
Diclofenac Potassium…………..50 mg

## PRODUCT DESCRIPTION

Brick red coloured, round, bioconvex, coated tablets.

## DESCRIPTION

Diclofenac potassium is chemically designated as 2-[(2,6-dichlorophenyl)amino]benzeneacetic acid, monopotassium salt. Its empirical formula is $C14H11CI2NO2K$ and its molecular weight is 334.25.

## PHARMACOLOGY

### Mechanism of Action

DIFNAL-K contains potassium salt of diclofenac, a non-steroidal compound with pronounced antirheumatic, analgesic, anti-inflammatory and antipyretic properties. Inhibition of prostaglandin biosynthesis is considered fundamental to its mechanism of action. Prostaglandins play an important role in causing inflammation, pain and fever.

Diclofenac has a rapid onset of action which makes them particularly suitable for the treatment of acute painful and inflammatory conditions.

Diclofenac *in vitro* does not suppress proteoglycan biosynthesis in cartilage at concentrations equivalent to the concentrations reached in humans.

### Pharmacodynamics (PD)

Diclofenac has been reported to exert a pronounced analgesic effect in moderate and severe pain. In the presence of inflammation, e.g. due to trauma or following surgical interventions, it rapidly relieves both spontaneous pain and pain on movement and diminishes inflammatory swelling and wound edema.

In primary dysmenorrhea the active substance is capable of relieving the pain and reducing the extent of bleeding.

In migraine attacks diclofenac has been reported to be effective in relieving the headache and in improving the accompanying symptoms nausea and vomiting.

**Pharmacokinetics**

*Absorption*

Diclofenac is completely absorbed from the gastrointestinal tract. The absorption sets in immediately after administration and the same amount is absorbed as from an equivalent dose of diclofenac sodium gastro-resistant tablets. Mean peak plasma concentrations of 3.8 micro mol/L are reported to be attained after 20 to 60 minutes after ingestion of one tablet of 50 mg. Ingestion together with food has no influence on the amount of diclofenac absorbed although onset and rate of absorption may be slightly delayed.

Since about half of diclofenac is metabolized during its first passage through the liver ("first pass" effect), the area under the concentration curve (AUC) is about half as large following oral or rectal administration as it is following a parenteral dose of equal size.

Pharmacokinetic behaviour does not change after repeated administration. No accumulation occurs provided the recommended dosage intervals are observed.

*Distribution*

Diclofenac binds to serum proteins (99.7%), mainly to albumin (99.4%). The apparent volume of distribution calculated is reported to be 0.12 to 0.17 L/kg.

Diclofenac enters the synovial fluid, where maximum concentrations are measured 2 to 4 hours after peak plasma values have been reached. The apparent half-life for elimination from the synovial fluid is reported to be 3 to 6 hours. Two hours after reaching peak plasma levels, concentrations of the active substance are already higher in the synovial fluid than in the plasma, and they remain higher for up to 12 hours.

Diclofenac has been reported in a low concentration (100 ng/mL) in breast milk in a nursing mother. The estimated amount ingested by an infant consuming breast milk is equivalent to a 0.03 mg/kg/day dose.

*Biotransformation/metabolism*

Biotransformation of diclofenac takes place partly by glucuronidation of the intact molecule, but mainly by single and multiple hydroxylation and methoxylation, resulting in several phenolic metabolites (3'-hydroxy-, 4'-hydroxy-, 5-hydroxy-, 4',5-dihydroxy-, and 3'-hydroxy-4'-methoxy- diclofenac), most of which are converted to glucuronide conjugates.

Two of these phenolic metabolites are biologically active, but to a much lesser extent than diclofenac.

*Elimination*

Total systemic clearance of diclofenac from plasma is reported to be 263 ± 56 mL/min (mean value ± SD). The terminal half-life in plasma is reported to be 1 to 2 hours. Four of the metabolites, including the two active ones, also have short plasma half-lives of 1 to 3 hours. One metabolite, 3'- hydroxy-4'-methoxy-diclofenac, has a much longer plasma half-life. However, this metabolite is virtually inactive.

About 60% of the administered dose is excreted in the urine as the glucuronide conjugate of the intact molecule and as metabolites, most of which are also converted to glucuronide conjugates. Less than 1% is excreted as unchanged substance. The rest of the dose is eliminated as metabolites through the bile in the faeces.

*Linearity/non-linearity*

The amount absorbed is in linear proportion to the size of the dose.

*Special populations*

- Geriatric patients: No relevant age-dependent differences in the drug's absorption, metabolism, or excretion have been reported.
- Renal impairment: In patients suffering from renal impairment, no accumulation of the unchanged active substance can be inferred from the single-dose kinetics when applying the usual dosage schedule. At a creatinine clearance of less than 10 mL/min, the calculated steady-state plasma levels of the hydroxy metabolites are reported to be about 4 times higher than in normal subjects. However, the metabolites are ultimately cleared through the bile.
- Hepatic impairment: In patients with chronic hepatitis or non-decompensated cirrhosis, the kinetics and metabolism of diclofenac are the same as in patients without liver disease.

## INDICATIONS

As short-term treatment for the following acute conditions in cases where particular importance is attached to a prompt onset of effect (within 30 minutes):

- Painful post-traumatic inflammatory states, e.g. due to sprains.
- Post-operative inflammation and pain e.g. following dental or orthopaedic surgery.
- Painful and/or inflammatory conditions in gynaecology e.g. primary dysmenorrhea or adnexitis.
- Acute migraine attacks. **DIFNAL-K** should not be used for migraine prophylaxis.
- Painful syndromes of the vertebral column (cervical/lumbar spondyloses)
- Non-articular rheumatism
- As an adjuvant in severe painful inflammatory infections of the throat, nose, or ears, e.g. pharyngotonsillitis, otitis. In keeping with general therapeutic principles, the underlying disease should be treated with basic therapy, as appropriate. Fever alone is not an indication.

## CONTRAINDICATIONS

**DIFNAL-K** is contraindicated in patients with

- Known hypersensitivity to diclofenac or any of the other excipients of the formulation
- Active gastric or intestinal ulcer, bleeding or perforation (see section **WARNINGS AND PRECAUTIONS** and also section **ADVERSE EFFECT**).
- Last trimester of pregnancy (see section **WOMEN OF CHILD-BEARING POTENTIAL, PREGNANCY**, **LACTATION** AND **FERTILITY**).
- Hepatic failure
- Renal failure (GFR <15 mL/min/1.73m$^2$).
- Severe cardiac failure (see section **WARNINGS AND PRECAUTIONS**)
- Established congestive heart failure (NYHA II-IV), ischemic heart disease, peripheral arterial disease and/or cerebrovascular disease.
- Like other non-steroidal anti-inflammatory drugs (NSAIDs), **DIFNAL-K** is also contraindicated in patients in whom the use of acetylsalicylic acid or other NSAIDs can precipitate asthma, angioedema, urticaria, or acute rhinitis (i.e. NSAID-induced cross-reactivity reactions) (see section **WARNINGS AND PRECAUTIONS** and also section **ADVERSE EFFECT**).

## DOSAGE AND ADMINISTRATION

**DIFNAL-K** tablets are available in the strength of 50 mg only and may not be suitable for all the dosing recommendations mentioned below. In such cases, other appropriate strengths/formulations of diclofenac must be used.

### Dosage

As a general recommendation, the dose should be individually adjusted. Adverse effects may be minimized by using the lowest effective dose for the shortest duration necessary to control symptoms (see section **WARNINGS AND PRECAUTIONS**).

### General target population:

### Adults

The recommended initial daily dose is 100 to 150 mg. In milder cases, 75 to 100 mg daily is usually sufficient. The total daily dose should generally be divided into 2 or 3 separate doses, as applicable.

In primary dysmenorrhea, the daily dose should be individually adjusted and is generally 50 to 150 mg. An initial dose of 50 mg is usually sufficient. If necessary, an initial dose of 100 mg can be prescribed with a maximum of 200 mg/day reached over the course of several menstrual cycles.

Treatment should be started on appearance of the first symptoms and, depending on the symptomatology, continued for a few days.

In migraine, an initial dose of 50 mg should be taken at the first signs of an impending attack. In cases where pain relief within 2 hours after the first dose is not sufficient, a further dose of 50 mg may be taken. If needed, further doses of 50 mg may be taken at intervals of 4 to 6 hours, not exceeding a total dose of 200 mg per day.

**Special populations**

*Pediatric patients (below 18 years of age)*
**DIFNAL-K** tablets are not recommended for use in children and adolescents below 14 years of age. For treatment in children and adolescents below 14 years of age, oral drops or suppositories of diclofenac 12.5 mg and 25 mg could be used. For adolescents aged 14 years and over, a daily dose of 75 to 100 mg is usually sufficient.

The maximum daily dose of 150 mg should not be exceeded. The total daily dose should generally be divided into 2 to 3 separate doses, as applicable.

The use of **DIFNAL-K** (all forms) in migraine attacks has not been reported in children and adolescents.

*Geriatric patients (aged 65 years or above)*
No adjustment of the starting dose is generally required for elderly patients. However, caution is indicated on basic medical grounds, especially for frail elderly patients or those with a low body weight (see section **WARNINGS AND PRECAUTIONS**).

*Established cardiovascular disease or significant cardiovascular risk factors*
Treatment with **DIFNAL-K** is generally not recommended in patients with established cardiovascular disease (congestive heart failure, established ischemic heart disease, peripheral arterial disease) or uncontrolled hypertension. If needed, patients with established cardiovascular disease, uncontrolled hypertension or significant risk factors for cardiovascular disease (e.g. hypertension, hyperlipidaemia, diabetes mellitus and smoking) should be treated with diclofenac only after careful consideration and only at doses ≤100 mg daily if treated for more than 4 weeks (see section **WARNINGS AND PRECAUTIONS**).

*Renal impairment*
**DIFNAL-K** is contraindicated in patients with renal failure (GFR <15 mL/min/1.73m$^2$) (see section **CONTRAINDICATIONS**).

There are no reported data in patients with renal impairment, therefore, no specific dose adjustment recommendations can be made. Caution is advised when administering **DIFNAL-K** to patients with renal impairment (see section **WARNINGS AND PRECAUTIONS**).

*<u>Hepatic impairment</u>*
**DIFNAL-K** is contraindicated in patients with hepatic failure (see section **CONTRAINDICATIONS**).

There are no reported data in patients with hepatic impairment, therefore, no specific dose adjustment recommendations can be made. Caution is advised when administering **DIFNAL-K** to patients with mild to moderate hepatic impairment (see section **WARNINGS AND PRECAUTIONS**).

***Method of administration***
The tablets should be swallowed whole with liquid, preferably before meals, and must not be divided or chewed.

## <u>WARNINGS AND PRECAUTIONS</u>

**General**
Caution is indicated in the elderly on basic medical grounds. In particular, it is recommended that the lowest effective dose be used in frail elderly patients or those with a low body weight.

As with other non-steroidal anti-inflammatory drugs including diclofenac, allergic reactions, including anaphylactic/anaphylactoid reactions, can also occur without earlier exposure to the drug. Hypersensitivity reactions can also progress to Kounis syndrome, a serious allergic reaction that can result in myocardial infarction. Presenting symptoms of such reactions can include chest pain occurring in association with an allergic reaction to diclofenac.

Like other NSAIDs, diclofenac may mask the signs and symptoms of the infection due to its pharmacodynamic properties.

**Cardiovascular Effects**
Treatment with NSAIDs including diclofenac, particularly at high dose and in long term, may be associated with an increased risk of serious cardiovascular events (including myocardial infarction and stroke).

Treatment with diclofenac is generally not recommended in patients with established cardiovascular disease (congestive heart failure (NYHA-I), established ischemic heart disease, peripheral arterial disease) or uncontrolled hypertension. If needed, patients with established cardiovascular disease, uncontrolled hypertension or significant risk factors for cardiovascular disease (e.g. hypertension, hyperlipidemia, diabetes mellitus and smoking) should be treated with diclofenac only after careful consideration and only at doses ≤100 mg daily when treatment continues for more than 4 weeks.

As the cardiovascular risks of diclofenac may increase with dose and duration of exposure, the lowest effective daily dose should be used for the shortest duration possible. The patient's need for symptomatic relief and response to therapy should be re-evaluated periodically, especially when treatment continues for more than 4 weeks.

Patients should remain alert for the signs and symptoms of serious arteriothrombotic events (e.g. chest pain, shortness of breath, weakness, slurring of speech), which can occur without warnings. Patients should be instructed to see a physician immediately in case of such an event.

An increased risk of arterial thrombotic events (for example myocardial infarction or stroke) associated with the use of diclofenac has been reported particularly at high dose (150 mg daily) and in long term treatment.

**Hematologic effects**
Use of diclofenac is recommended only for short-term treatment. If, however, diclofenac is used for a prolonged period, monitoring of the blood count is recommended, as with other NSAIDs.

Like other NSAIDs, diclofenac may temporarily inhibit platelet aggregation. Patients with defects of hemostasis should be carefully monitored.

**Respiratory effects (Pre-existing asthma)**
In patients with asthma, seasonal allergic rhinitis, swelling of the nasal mucosa (i.e. nasal polyps), chronic obstructive pulmonary diseases or chronic infections of the respiratory tract (especially if linked to allergic rhinitis-like symptoms), reactions on NSAIDs like asthma exacerbations (so-called intolerance to analgesics/analgesics-asthma), Quincke's edema or urticaria are more frequent than in other patients. Therefore, special caution is recommended in such patients (readiness for emergency). This is applicable as well for patients who are allergic to other substances, e.g. with skin reactions, pruritus or urticaria.

Like other drugs that inhibit prostaglandin synthetase activity, diclofenac sodium and other NSAIDs can precipitate bronchospasm if administered to patients suffering from, or with a previous history of bronchial asthma.

**Hepatobiliary effects**
Close medical surveillance is required when prescribing diclofenac to patients with impaired hepatic function, as their condition may be exacerbated.

As with other NSAIDs, including diclofenac, values of one or more liver enzymes may increase. During prolonged treatment with diclofenac, regular monitoring of hepatic function is indicated as a precautionary measure. If abnormal liver function tests persist or worsen, if clinical signs or symptoms consistent with liver disease develop, or if other manifestations occur (e.g. eosinophilia, rash), diclofenac should be discontinued. Hepatitis may occur with use of diclofenac without prodromal symptoms.

Caution is called for when using diclofenac in patients with hepatic porphyria, since it may trigger an attack.

**Gastrointestinal Effects**

Gastrointestinal bleeding, ulceration or perforation, which can be fatal, have been reported with all NSAIDs, including diclofenac, and may occur at any time during treatment, with or without warning symptoms or a previous history of serious gastrointestinal events. They generally have more serious consequences in the elderly. If gastrointestinal bleeding or ulceration occurs in patients receiving diclofenac, the treatment should be discontinued.

As with all NSAIDs, including diclofenac, close medical surveillance is imperative and particular caution should be exercised when prescribing diclofenac in patients with symptoms indicative of gastrointestinal (GI) disorders or with a history suggestive of gastric or intestinal ulceration, bleeding or perforation (see section **ADVERSE EFFECT**). The risk of GI bleeding is higher with increasing NSAID doses and in patients with a history of ulcer, particularly if complicated with hemorrhage or perforation and in the elderly.

To reduce the risk of GI toxicity in patients with a history of ulcer, particularly if complicated with hemorrhage or perforation, and in the elderly, the treatment should be initiated and maintained at the lowest effective dose.

Combination therapy with protective agents (e.g. proton pump inhibitors or misoprostol) should be considered for these patients, and also for patients requiring concomitant use of low-dose acetylsalicylic acid (ASA) or other drugs likely to increase gastrointestinal risk.

Patients with a history of GI toxicity, particularly the elderly, should report any unusual abdominal symptoms (especially GI bleeding). Caution is recommended in patients receiving concomitant medications which could increase the risk of ulceration or bleeding, such as systemic corticosteroids, anticoagulants, anti-platelet agents or selective serotonin-reuptake inhibitors (see section **DRUG INTERACTIONS**).

Close medical surveillance and caution should also be exercised in patients with ulcerative colitis or Crohn’s disease, as their condition may be exacerbated (see section **ADVERSE EFFECT**).

NSAIDs, including diclofenac, may be associated with increased risk of gastro-intestinal anastomotic leak. Close medical surveillance and caution are recommended when using diclofenac after gastro-intestinal surgery.

**Severe Skin Reactions**

Serious skin reactions, some of them fatal, including exfoliative dermatitis, Stevens-Johnson syndrome and toxic epidermal necrolysis, have been reported very rarely in association with the use of NSAIDs, including diclofenac (see section **ADVERSE EFFECT**). Patients appear to be at highest risk of these reactions early in the course of therapy, the onset of the reaction occurring in the majority of cases within the first month of treatment. Diclofenac should be discontinued at the first appearance of skin rash, mucosal lesions or any other sign of hypersensitivity.

As with other NSAIDs, allergic reactions, including anaphylactic/anaphylactoid reactions, can also occur in rare cases with diclofenac without earlier exposure to the drug.

**SLE and mixed connective tissue disease**
In patients with systemic lupus erythematosus (SLE) and mixed connective tissue disorders there may be an increased risk of aseptic meningitis.

**Renal effects**
As fluid retention and edema have been reported in association with NSAID therapy, including diclofenac, particular caution is called for in patients with impaired cardiac or renal function, history of hypertension, the elderly, patients receiving concomitant treatment with diuretics or medicinal products that can significantly impact renal function, and in those patients with substantial extracellular volume depletion from any cause, e.g. before or after major surgery (see section **CONTRAINDICATIONS**). Monitoring of renal function is recommended as a precautionary measure when using diclofenac in such cases. Discontinuation of therapy is usually followed by recovery to the pre-treatment state.

**Geriatric patients**
Caution is indicated in the elderly on basic medical grounds, especially in frail elderly patients or those with a low body weight.

**Interactions with NSAIDs**
The concomitant use of diclofenac with systemic NSAIDs including cyclooxygenase-2 selective inhibitors, should be avoided due to the potential for additive undesirable effects (see section **DRUG INTERACTIONS**).

**Masking signs of infections**
Like other NSAIDs, diclofenac may mask the signs and symptoms of infection due to its pharmacodynamic properties.

**Effects on ability to drive and use machines**
Patients who experience visual disturbances, dizziness, vertigo, somnolence, central nervous system disturbances, drowsiness, or fatigue while taking NSAIDs should refrain from driving or operating machinery.

**WARNINGS**

**Risk of GI ulceration, bleeding and perforation with NSAIDS**

Serious GI toxicity such as bleeding, ulceration and perforation can occur at any time, with or without warning symptoms, in patients treated with NSAID therapy. Although minor upper GI problems (e.g dyspepsia) are common, usually developing early in therapy, prescribers should remain alert for ulceration and bleeding in patients treated with NSAIDs even in the absence of previous GI tract symptoms.

Studies to date have not identified any subset of patients not at risk of developing peptic ulceration and bleeding. Patients with prior history of serious adverse events and other risk factors associated with peptic ulcer disease (e.g. alcoholism, smoking, and corticosteroid therapy) are at increased risk. Elderly or debilitated patients seem to tolerate ulceration or bleeding less than other individuals and account for most spontaneous reports for fatal GI events.

**PREGNANCY**

There are insufficient data on the use of diclofenac in pregnant women. Increased risk of miscarriage and or cardiac malformation and gastroschisis has been reported after use of a prostaglandin synthesis inhibitor (such as NSAIDs) in early pregnancy, however the overall data are inconclusive. The risk is believed to increase with dose and duration of therapy. Administration of a prostaglandin synthesis inhibitor has been reported to increase pre-and post-implantation loss and embryo-foetal lethality in animals. Diclofenac should not be used during the first two trimesters of pregnancy unless the expected benefits to the mother outweigh the risks to the fetus. As with other NSAIDs, use of diclofenac during the third trimester of pregnancy is contraindicated owing to the possibility of uterine inertia, fetal renal impairment with subsequent oligohydramnios and/or premature closure of the ductus arteriosus (see section **CONTRAINDICATIONS** and also section **PRECLINICAL SAFETY DATA**).

Increased incidences of various malformations, including cardiovascular, have been reported in animals given a prostaglandin synthesis inhibitor during organogenetic period. During the third trimester of pregnancy, all prostaglandin synthesis inhibitors may expose the foetus to:

- cardiopulmonary toxicity (with premature closure of the ductus arteriosus and pulmonary hypertension
- renal dysfunction, which may progress to renal failure with oligo-hydroamniosis

The mother and the neonate, at the end of the pregnancy, to:

- possible prolongation of bleeding time, an anti-aggregating effect which may occur even at very low doses
- inhibition of uterine contractions resulting in delayed or prolonged labour.

Consequently, diclofenac is contraindicated during the third trimester of pregnancy.

**LACTATION**

Diclofenac is excreted in breast milk in small amounts. Therefore, diclofenac should not be administered during breast-feeding in order to avoid undesirable effects in the infant.

**FERTILITY**

As with other NSAIDs, the use of diclofenac may impair female fertility and is not recommended in women attempting to conceive. In women who have difficulties conceiving or who are undergoing investigation of infertility, withdrawal of diclofenac should be considered.

**WOMEN OF CHILD-BEARING POTENTIAL**

There are no reported data to suggest any recommendations for women of child-bearing potential.

**PRECLINICAL SAFETY DATA**

The reported preclinical data reveal no specific hazard for humans at the intended therapeutic doses. Diclofenac has not been reported to be teratogenic in mice, rats or rabbits.

Diclofenac has no influence on the fertility of parent animals in rats. Except for minimal fetal effects at maternally toxic doses, the prenatal, perinatal and postnatal development of the offspring have not been reported to be affected.

Administration of NSAIDs (including diclofenac) have been reported to inhibit ovulation in the rabbit and implantation and placentation in the rat, and led to premature closure of the ductus arteriosus in the pregnant rat. Maternally toxic doses of diclofenac have been reported to be associated with dystocia, prolonged gestation, decreased fetal survival, and intrauterine growth retardation in rats. The slight effects of diclofenac on reproduction parameters and delivery as well as constriction of the ductus arteriosus *in utero* are pharmacologic consequences of this class of prostaglandin synthesis inhibitors (see section **CONTRAINDICATIONS** and also section **WOMEN OF CHILD-BEARING POTENTIAL, PREGNANCY, LACTATION** and **FERTILITY**).

**DRUG INTERACTIONS**

The following interactions include those reported with diclofenac sugar-coated tablets and/or other pharmaceutical forms of diclofenac.

***Observed interactions to be considered***

*CYP2C9 inhibitors*: Caution is recommended when co-prescribing diclofenac with CYP2C9 inhibitors (such as voriconazole), which could result in a significant increase in peak plasma concentrations and exposure to diclofenac.

*Lithium*: If used concomitantly, diclofenac may raise plasma concentrations of lithium. Monitoring of the serum lithium level is recommended.

*Digoxin*: If used concomitantly, diclofenac may raise plasma concentrations of digoxin. Monitoring of the serum digoxin level is recommended.

*Diuretics and antihypertensive agents*: Like other NSAIDs, concomitant use of diclofenac with diuretics or antihypertensive agents (e.g. beta-blockers, angiotensin converting enzyme (ACE) inhibitors) may cause a decrease in their antihypertensive effect. Therefore, the combination should be administered with caution and patients, especially the elderly, should have their blood pressure periodically monitored. Patients should be adequately hydrated and consideration should be given to monitoring of renal function after initiation of concomitant therapy and periodically thereafter, particularly for diuretics and ACE inhibitors due to the increased risk of nephrotoxicity (see section **WARNINGS AND PRECAUTIONS**).

*Ciclosporin and tacrolimus*: Diclofenac, like other NSAIDs, may increase the nephrotoxicity of ciclosporin and tacrolimus due to the effect on renal prostaglandins. Therefore, it should be given at doses lower than those that would be used in patients not receiving ciclosporin or tacrolimus.

*Drugs known to cause hyperkalemia*: Concomitant treatment with potassium-sparing diuretics, ciclosporin, tacrolimus or trimethoprim may be associated with increased serum potassium levels, which should therefore be monitored frequently (see section **WARNINGS AND PRECAUTIONS**).

*Quinolone antibacterials*: There have been isolated reports of convulsions which may have been due to concomitant use of quinolones and NSAIDs. Therefore, caution should be exercised when considering the use of a quinolone in patients who are already receiving an NSAID.

***Anticipated interactions to be considered***

*Other NSAIDs and corticosteroids*: Concomitant administration of diclofenac and other systemic NSAIDs or corticosteroids may increase the frequency of gastrointestinal undesirable effects (see section **WARNINGS AND PRECAUTIONS**). Avoid concomitant use of two or more NSAIDs.

*Anticoagulants and anti-platelet agents*: Caution is recommended since concomitant administration could increase the risk of bleeding (see section **WARNINGS AND PRECAUTIONS**). Although clinical investigations do not appear to indicate that diclofenac affects the action of anticoagulants, there are reports of an increased risk of hemorrhage in patients receiving diclofenac and anticoagulants concomitantly. Close monitoring of such patients is therefore recommended.

*Selective serotonin reuptake inhibitors (SSRIs)*: Concomitant administration of systemic NSAIDs, including diclofenac, and SSRIs may increase the risk of gastrointestinal bleeding (see section **WARNINGS AND PRECAUTIONS**).

*Antidiabetics*: Diclofenac can be given together with oral antidiabetic agents without influencing their clinical effect. However, there have been isolated reports of both hypoglycemic and hyperglycemic effects necessitating changes in the dosage of the antidiabetic agents during treatment with diclofenac. For this reason, monitoring of the blood glucose level is recommended as a precautionary measure during concomitant therapy.

There have also been isolated reports of metabolic acidosis when diclofenac was co-administered with metformin, especially in patients with pre-existing renal impairment.

*Phenytoin*: When using phenytoin concomitantly with diclofenac, monitoring of phenytoin plasma concentrations is recommended due to an expected increase in exposure to phenytoin.

*Methotrexate*: Diclofenac can inhibit the tubular renal clearance of methotrexate hereby increasing methotrexate levels. Caution is recommended when NSAIDs, including diclofenac, are administered less than 24 hours before or after treatment with methotrexate, since blood concentrations of methotrexate may rise and the toxicity of this substance be increased. Cases of serious toxicity have been reported when methotrexate and NSAIDs, including diclofenac are given within 24 hours of each other. This interaction is mediated through accumulation of methotrexate resulting from impairment of renal excretion in the presence of the NSAID.

*CYP2C9 inducers*: Caution is recommended when co-prescribing diclofenac with CYP2C9 inducers (such as rifampicin), which could result in a significant decrease in plasma concentration and exposure to diclofenac.

*Colestipol and cholestyramine*: These agents can induce a delay or decrease in absorption of diclofenac. Therefore, it is recommended to administer diclofenac at least one hour before or 4 to 6 hours after administration of colestipol/ cholestyramine.

*Cardiac glycosides*: Concomitant use of cardiac glycosides and NSAIDs in patients may exacerbate cardiac failure, reduce GFR and increase plasma glycoside levels.

*Mifepristone*: NSAIDs should not be used for 8-12 days after mifepristone administration as NSAIDs can reduce the effect of mifepristone.

## ADVERSE EFFECT

| | |
|---|---|
| **Blood and lymphatic system disorders** | |
| Very rare: | Thrombocytopenia, leukopenia, anemia (including hemolytic and aplastic anemia), agranulocytosis |
| **Immune system disorders** | |
| Rare: | Hypersensitivity, anaphylactic and anaphylactoid reactions (including hypotension and shock) |
| Very rare: | Angioedema (including face edema) |
| **Psychiatric disorders** | |
| Very rare: | Disorientation, depression, insomnia, nightmare, irritability, psychotic disorder |
| **Nervous system disorders** | |
| Common: | Headache, dizziness |
| Rare: | Somnolence, tiredness |
| Very rare: | Paresthesia, memory impairment, convulsion, anxiety, tremor, aseptic |
| Unknown | Confusion, hallucinations, disturbances of sensation, malaise |
| **Eye disorders** | |
| Very rare: | Visual impairment, blurred vision, diplopia |
| Unknown | Optic neuritis |
| **Ear and labyrinth disorders** | |
| Common: | Vertigo |
| Very rare: | Tinnitus, impaired hearing |
| **Cardiac disorders** | |
| Uncommon*: | Myocardial infarction, cardiac failure, palpitations, chest pain |
| Frequency not known | Kounis syndrome |
| **Vascular disorders** | |
| Very rare: | Hypertension, vasculitis, hypotension |
| **Respiratory, thoracic and mediastinal disorders** | |
| Rare: | Asthma (including dyspnea) |
| Very rare: | Pneumonitis |
| **Gastrointestinal disorders** | |
| Common: | Nausea, vomiting, diarrhea, dyspepsia, abdominal pain, flatulence, decreased appetite |
| Rare: | Gastritis, gastrointestinal hemorrhage, hematemesis, hemorrhagic diarrhea, melena, gastrointestinal ulcer (with or without bleeding, gastrointestinal stenosis, or perforation, which may lead to peritonitis) |

| | |
|---|---|
| Very rare: | Colitis (including hemorrhagic colitis, ischemic colitis and exacerbation of ulcerative colitis or Crohn's disease), constipation, stomatitis (including ulcerative stomatitis), glossitis, esophageal disorder, intestinal diaphragm disease, pancreatitis |
| Unknown | Ischaemic colitis |
| **Hepatobiliary disorders** | |
| Common: | Transaminases increased |
| Rare: | Hepatitis, jaundice, liver disorder |
| Very rare: | Fulminant hepatitis, hepatic necrosis, hepatic failure |
| **Skin (dermatological) and subcutaneous tissue disorders** | |
| Common: | Rash |
| Rare: | Urticaria |
| Very rare: | Bullous dermatitis, eczema, erythema, erythema multiforme, Stevens-Johnson syndrome, toxic epidermal necrolysis (Lyell's syndrome), exfoliative dermatitis, alopecia, photosensitivity reaction, purpura, Henoch-Schoenlein purpura, pruritus |
| **Renal and urinary disorders** | |
| Very rare: | Acute kidney injury (acute renal failure), hematuria, proteinuria, nephrotic syndrome, tubulointerstitial nephritis, renal papillary necrosis |
| **Reproductive system and breast disorders** | |
| Very rare | Impotence |
| **General disorders and administration site conditions** | |
| Rare | Edema |

***The frequency reflects reported data from long-term treatment with a high dose (150 mg/day***

**Description of selected adverse drug reactions**

*Arteriothrombotic events*

Meta-analysis and pharmacoepidemiological data point towards an increased risk of arteriothrombotic events (for example myocardial infarction) associated with the use of diclofenac, particularly at a high dose (150 mg daily) and during long-term treatment. (see section **WARNINGS AND PRECAUTIONS**).

*Visual effects*

Visual disturbances such as visual impairment, blurred vision or diplopia appear to be NSAID class effects and are usually reversible on discontinuation. A likely mechanism for the visual disturbances is the inhibition of prostaglandin synthesis and other related compounds that alter the regulation of retinal blood flow resulting in potential changes in vision. If such symptoms occur during diclofenac treatment, an ophthalmological examination may be considered to exclude other causes.

## OVERDOSAGE

**Symptoms**
There is no typical clinical picture resulting from diclofenac overdosage. Overdosage can cause symptoms such as headache, nausea, vomiting, epigastric pain, gastrointestinal hemorrhage, diarrhea, dizziness, disorientation, excitation, coma, drowsiness, tinnitus, fainting or convulsions. In the event of significant poisoning, acute renal failure and liver damage are possible.

**Therapeutic measures**
Management of acute poisoning with NSAIDs, including diclofenac, essentially consists of supportive measures and symptomatic treatment. Supportive measures and symptomatic treatment should be given for complications such as hypotension, renal failure, convulsions, gastrointestinal disorder, and respiratory depression.

Special measures such as forced diuresis, dialysis or hemoperfusion are probably of no help in eliminating NSAIDs, including diclofenac, due to the high protein binding and extensive metabolism. Activated charcoal may be considered after ingestion of a potentially toxic overdose, and gastric decontamination (e.g. vomiting, gastric lavage) after ingestion of a potentially life-threatening overdose.

## STORAGE
Store below 30°C, protected from moisture

**KEEP ALL MEDICINES OUT OF REACH OF CHILDREN**

## PACKING
Strip of 10 x 10’s, 100 x 10’s

**Product Registration Holder/ Manufactured by,**
RANBAXY (Malaysia) Sdn. Bhd,
Lot 23, Bakar Arang Industrial Estate
08000 Sungai Petani, Kedah, Malaysia.

Date of revision: January 2025